STAT 4001

Topics in Data Science A

FITFR Analysis Report

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Introduction

The data were generously supplied to us from Karl Berator. There were 100 total subjects and 58 variables recorded. The primary variables of interest included:

- `treatment`: treatment group assignment. Factor with three levels A, B, and C
- `age`: age of participant (years). Min = 25, Max = 45
- `sex`: sex of participant (female/male)
- `smoker`: whether participant is smoker or not
- `51 SNPs`: genetic markers that we can use as genetic covariates
- `AF`: continous outcome variable. Abbreviation stands for "activating factor". Min = 26.02, Max = 95.99
- `PR`: binary outcome variable. Abbreviation stands for "positive response". Factor with levels yes and no.

Our research question involved determining whether treatment group was significantly associated with AF and PR levels while adjusting for potential confounding effects of age, sex, smoking status, and genetic variation.

Identifying which treatment most effectively influences AF and PR, and whether this effect is attributable to treatment itself rather than demographic or genetic differences between groups, has direct implications for treatment selection and the design of future studies.

Methods

Data Cleaning and Pre-processing

The raw data file contained a two-row header structure, with the first row containing broad category labels and the second containing variable names; the first row was skipped on import. The last four rows of the file were artefacts (three empty rows and one garbled header fragment) and were removed by filtering on non-missing participant ID, as all genuine participants have a valid ID.

Upon inspection, 56 of 58 columns were read in as character type due to the Excel format; only `ID` and `AF` were numeric. The following type conversions were applied:

- `ID` and `age` were coerced to integer and numeric respectively; `AF` was confirmed to be numeric.
- `sex` was recoded to be consistent Female/Male (correcting capitalisation and abbreviation of "M") and then factorised.
- `treatment` was uppercased (from a/b/c to A/B/C) and then factorised with reference level A.
- `PR` was recoded from yes/no strings to a binary factor (0 = no, 1 = yes).

Two participants had `age = -99`, a sentinel value indicating missing data; these were recoded to `NA`. These two observations were excluded from model fitting, leaving an analytic sample of 98 participants. No other variables contained missing values.

Two zero-variance variables were identified and excluded from all models: `smoker` (all participants were non-smokers) and `rs123456` (monomorphic across the sample, all participants carrying the CC genotype). Both variables were retained in the cleaned dataset for transparency.

The remaining 50 SNP variables were encoded under a standard additive genetic model. For each SNP, the reference allele was determined as the allele present in the most frequent homozygous genotype. Each participant's genotype was then recoded as the count of alternate alleles carried (0 = homozygous reference, 1 = heterozygous, 2 = homozygous alternate). Encoded SNP values were stored as ordered factors with levels $0 < 1 < 2$.

Statistical Models

Two separate models were fitted to address the two outcomes of interest.

Activating factor (AF) was modelled using multiple linear regression. Linear regression was chosen as AF is a continuous variable with no natural bounds in this context, and the distribution of AF within treatment groups showed no severe skew. The full model takes the form:

$$AF_i = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + \beta_4 \text{SNP}_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

where β_0 is the intercept, β_1, β_2, β_3, and β_4 are the coefficients for treatment, age, sex, and SNP dosage respectively, and ϵ_i is the residual error assumed to be independently and normally distributed with mean zero and constant variance σ^2. The baseline and main effects models are nested special cases, obtained by setting $\beta_4 = 0$ and $\beta_2 = \beta_3 = \beta_4 = 0$ respectively. Coefficients can be interpreted as differences in AF units themselves. Robust regression was explored but found unnecessary due to a very high model fit (adjusted $R^2 = 0.977$). Log transformation of AF was also considered but not applied, as the within-group distributions were approximately symmetric.

Positive response (PR) was modelled using logistic regression. Logistic regression was chosen as PR is a binary outcome. Standard linear regression was considered but rejected as it can produce predicted probabilities outside [0, 1] and assumes linearity on the raw scale. The full model takes the form:

$$\log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + \beta_4 \text{SNP}_i$$

where $p_i = P(\text{PR}_i = \text{yes})$ is the probability of a positive response for participant i, and the left-hand side is the log-odds of PR. As with the AF model, the baseline and main effects models are nested special cases obtained by setting $\beta_4 = 0$ and $\beta_2 = \beta_3 = \beta_4 = 0$ respectively. Coefficients are exponentiated to produce odds ratios, representing the multiplicative change in the odds of PR per unit increase in each predictor. Probit regression was also considered; as it yields conclusions consistent with logistic regression in practice, logistic regression was preferred for its more interpretable odds ratio parameterisation.

Both models followed the same nested structure, with each subsequent model adding a further set of predictors to assess their incremental contribution:

1. **Baseline model**: treatment only.
2. **Main effects model**: treatment + age + sex.
3. **SNP-augmented model**: treatment + age + sex + top SNP candidate(s).

Smoking status and `rs123456` were excluded from all models as zero-variance variables.

Model Selection and Tuning

For each outcome, the three nested models were compared sequentially, where each successive model contains all predictors from the previous model plus an additional set of predictors.

AF: Models were compared using AIC and adjusted R^2, with ANOVA F-tests used to formally assess whether each additional set of predictors significantly improved model fit. A difference in AIC of less than 2 between successive models was treated as negligible evidence in favour of the more complex model, in which case the simpler model was preferred.

PR: Models were compared using AIC and McFadden's pseudo-R^2, with likelihood ratio tests used to formally assess improvement in fit. The same AIC decision rule was applied.

Assumption Checking

For the **AF linear regression model**, four standard diagnostics were examined: a residuals vs fitted plot (linearity and homoscedasticity), a Q-Q plot (normality of residuals), a scale-location plot (constant variance), and Cook's distance (influential observations). Variance inflation factors (VIF) were computed to assess multicollinearity, with $\text{GVIF}\hat{}(1/(2 \cdot \text{Df})) > 2$ used as the threshold for concern.

For the **PR logistic regression model**, the following diagnostics were applied:

- **Linearity**: Box-Tidwell test, assessing whether the log-odds of PR is linear in age via an age × log(age) interaction term.

- **Complete separation**: cross-tabulations of each categorical predictor against PR, with any zero cell count indicating separation.
- **Goodness of fit**: Hosmer-Lemeshow test (g = 10 groups), with $p > 0.05$ indicating adequate fit.
- **Multicollinearity**: VIF as above.
- **Influential observations**: Cook's distance with threshold 4/n.

SNP Candidate Selection

With 100 participants and 50 SNPs, simultaneously including all SNPs would violate the events per variable (EPV) rule of thumb, which recommends a minimum of 10 observations per predictor for stable regression estimates (Peduzzi et al. 1996).

Two approaches were considered for reducing the dimensionality of the SNP data prior to modelling.

Principal component analysis (PCA) was explored as a means of summarising genetic variation into a smaller number of uncorrelated components; however, this approach was not carried forward as the resulting components are linear combinations of all SNPs and cannot be interpreted in terms of individual genetic variants, which is of primary interest here.

Univariable regression screening was therefore adopted instead, as it places each SNP on the same inferential scale as the final model and retains the direct interpretability of individual SNP effects. For AF, a separate univariable linear regression was fitted for each SNP; for PR, a separate univariable logistic regression. Two thresholds were assessed: the Bonferroni-corrected threshold ($p < 0.05/50 = 0.001$)which accounts for the inflated false positive rate from running 50 simultaneous tests, and the nominal threshold ($p < 0.05$). As no SNPs survived Bonferroni correction (expected given $n = 100$), SNPs passing the nominal threshold were carried forward as exploratory candidates.

Univariable regression screening was therefore adopted instead, as it places each SNP on the same inferential scale as the final model and retains the direct interpretability of individual SNP effects. For each of the 50 SNPs, a separate univariable model was fitted with SNP dosage (0, 1, 2) as the sole predictor: a linear regression for AF and a logistic regression for PR. The p-value for the SNP coefficient in each model was extracted, representing the probability of observing an association of that magnitude or greater under the null hypothesis of no SNP effect.

Two thresholds were assessed:

- A SNP was considered to pass the Bonferroni-corrected threshold if its p-value was less than 0.001, calculated as 0.05 divided by the total number of tests to control the familywise error rate across all simultaneous comparisons.
- A SNP was considered to pass the nominal threshold if its p-value was less than 0.05.

SNPs were ranked by p-value and those passing the nominal threshold were carried forward into the multivariable models as exploratory candidates. As no SNPs survived Bonferroni correction, which is expected given the limited statistical power with only 100 participants, all SNP findings are treated as preliminary and should be interpreted with caution. SNPs were ranked by p-value and those passing the nominal threshold were carried forward into the multivariable models as exploratory candidates. As no SNPs survived Bonferroni correction, SNPs passing the nominal threshold were carried forward as exploratory candidates.

Software

All analyses were conducted in R version 4.5.2 (R Core Team 2025). The following packages were used: tidyverse version 2.0.0 (Wickham et al. 2019) for data manipulation and visualisation, readxl version 1.4.5 (Wickham and Bryan 2025) for data import, table1 (Rich 2025) for descriptive statistics, patchwork (Pedersen 2025) for figure composition, scales (Wickham, Pedersen, and Seidel 2025) for axis formatting, broom version 1.0.12 (Robinson et al. 2026) for tidy model output, ResourceSelection version 0.3-6 (Lele, Keim, and Solymos 2023) for the Hosmer-Lemeshow test, car version 3.1-5 (Fox and Weisberg 2019) for variance inflation factors, and kableExtra (Zhu 2024) for table formatting. Package versions are reported in the session information appended to this document.

Result

Descriptive Statistics

Table 1 summarises participant characteristics stratified by treatment group. The three groups were reasonably balanced in sex distribution and unadjusted PR rate, but differed significantly in AF ($p < 0.001$) and marginally in age ($p = 0.040$). Group B had the highest mean AF at 76.93 (SD = 11.22), followed by group C at 69.24 (SD = 13.00) and group A at 37.53 (SD = 6.37), with the gap between group A and the other two groups being particularly pronounced. Group B was also slightly older on average at 36.32 years, compared to 33.09 and 33.67 years for groups A and C respectively.

Table 1: Descriptive statistics stratified by treatment group (A, B, C) for 100 participants. Continuous variables are presented as mean (SD); categorical variables as n (%). Treatment groups differ significantly in AF ($p < 0.001$) and marginally in age ($p = 0.040$), but not in sex or PR distribution.

	A (N=32)	**B (N=29)**	**C (N=39)**	**Overall (N=100)**
age				
Mean (SD)	34.3 (6.12)	36.3 (5.68)	32.4 (6.34)	34.2 (6.23)
Median [Min, Max]	33.5 [25.0, 45.0]	37.0 [25.0, 45.0]	30.0 [25.0, 45.0]	34.5 [25.0, 45.0]
Missing	0 (0%)	1 (3.4%)	1 (2.6%)	2 (2.0%)
sex				
Female	15 (46.9%)	17 (58.6%)	19 (48.7%)	51 (51.0%)
Male	17 (53.1%)	12 (41.4%)	20 (51.3%)	49 (49.0%)
AF				
Mean (SD)	37.5 (6.37)	76.9 (11.2)	69.2 (13.0)	61.3 (19.8)
Median [Min, Max]	36.4 [26.0, 48.0]	77.8 [53.9, 96.0]	64.3 [52.8, 95.4]	61.1 [26.0, 96.0]
PR				
no	15 (46.9%)	10 (34.5%)	24 (61.5%)	49 (49.0%)
yes	17 (53.1%)	19 (65.5%)	15 (38.5%)	51 (51.0%)

The distribution of continuous variables is shown in Figure 1. AF was approximately uniformly distributed across its observed range of 26.02 to 95.99, with a mean of 61.32 (SD = 19.8) and no severe skew or obvious multimodality. Age ranged from 25 to 45 years with a mean of 34.2 (SD = 6.2); two observations originally recorded as the sentinel value -99 were recoded to missing prior to analysis. Among the categorical variables (Figure 2), treatment group C had the largest number of participants at 39, followed by A at 32 and B at 29. Sex was nearly balanced at 51 female and 49 male participants, and the PR outcome was evenly split between positive and negative responses (51 vs 49), indicating no class imbalance that would complicate binary modelling.

Primary Relationships

The association between treatment group and AF is displayed in Figure 3. AF differed substantially across the three groups (Kruskal-Wallis $p < 0.001$), with group A showing a markedly lower median AF of approximately 37 compared to approximately 78 for group B and 64 for group C. The interquartile ranges of group A and the other two groups did not overlap, and the within-group spread was considerably narrower for group A, together suggesting a strong and consistent treatment effect on AF. The unadjusted PR rates also varied across groups (Figure 4): group B had the highest positive response rate at 65.5%, followed by group A at 53.1% and group C at 38.5%. This difference did not reach statistical significance at the 5% level (^{2} p =

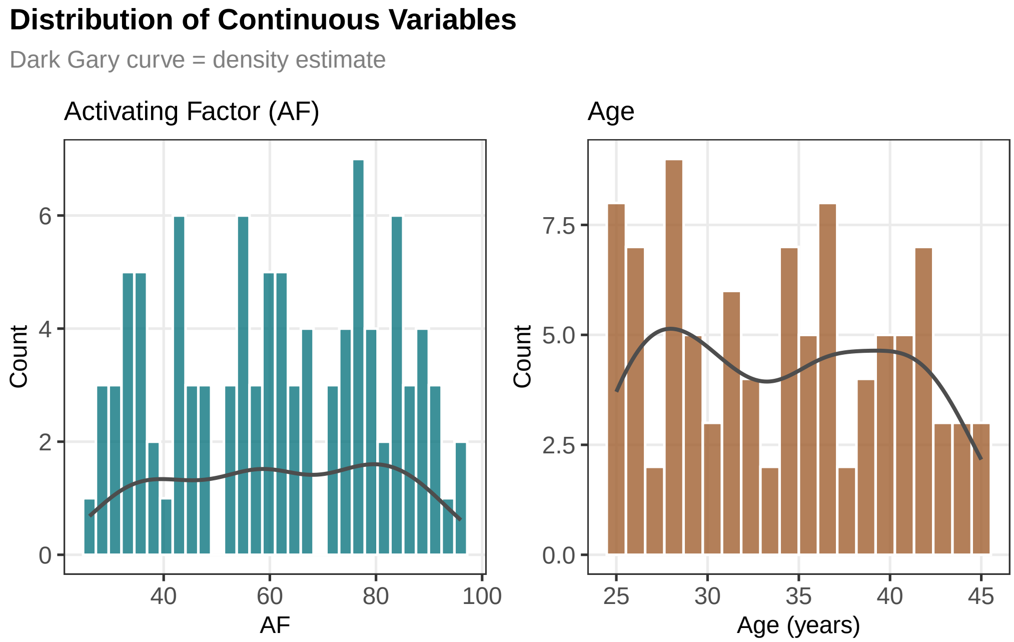

Figure 1: Distribution of continuous variables: activating factor (AF, left) and age (right). AF ranges from 26.02 to 95.99 (mean = 61.32, SD = 19.8) and is approximately uniformly distributed with no severe skew. Age ranges from 25 to 45 years (mean = 34.2, SD = 6.2) with two observations recoded from sentinel value -99 to missing prior to plotting.

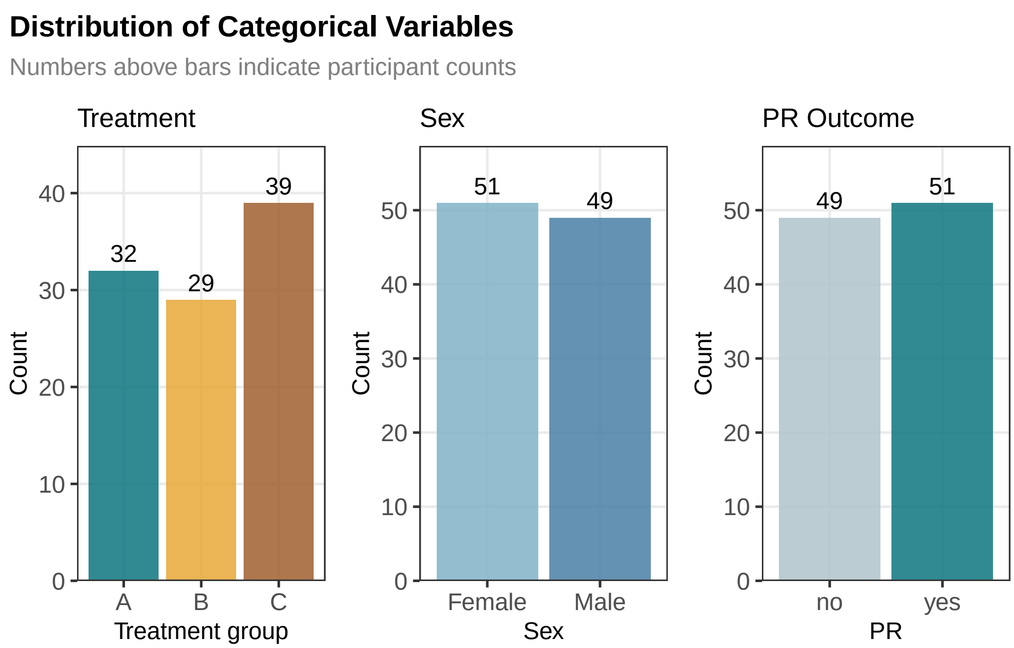

Figure 2: Distribution of categorical variables: treatment group (left), sex (centre), and PR outcome (right). Treatment group C has the largest number of participants (n = 39), followed by A (n = 32) and B (n = 29). Sex is approximately balanced (Female: n = 51, Male: n = 49). PR outcome is evenly split (no: n = 49, yes: n = 51), indicating no class imbalance for binary modelling.

0.084), and the pattern — group C having the second highest AF but the lowest PR rate — suggests the relationship between treatment, AF, and PR is not straightforward.

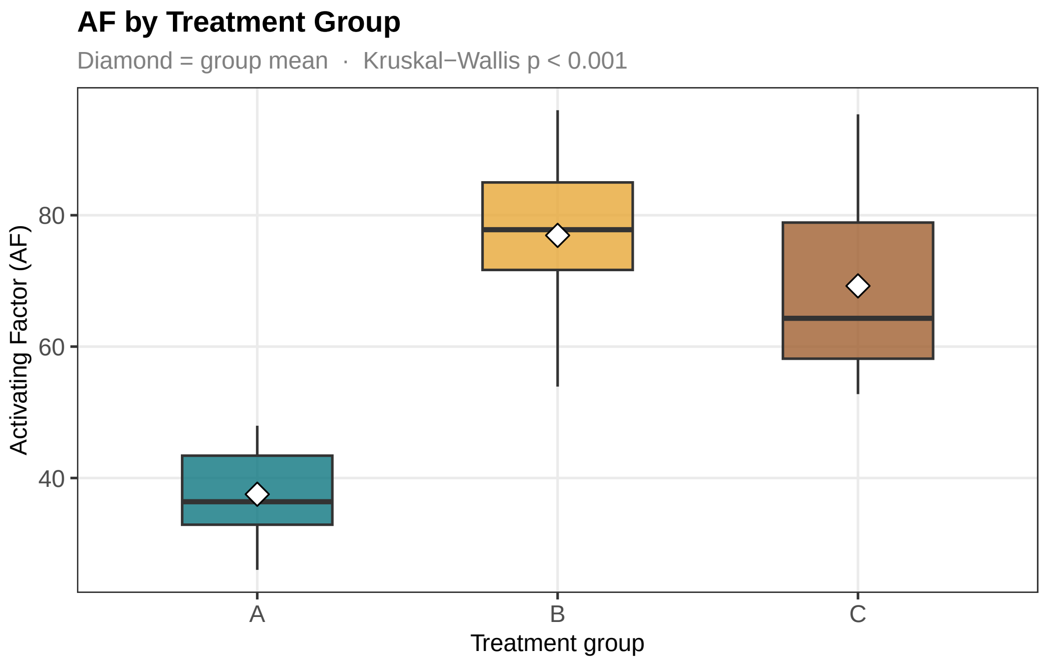

Figure 3: Boxplot showing the activating factor (AF) by treatment group (A, B, C) among 100 participants. Boxes show the interquartile range, horizontal line in box shows the median, and diamonds indicate the mean for each group. Treatment group A has a markedly lower AF (median ~37) compared to groups B (median ~78) and C (median ~64) and their interquartile ranges do not overlap, suggesting a strong treatment effect. Kruskal-Wallis $p < 0.001$.

The association between age and both outcomes is shown in Figure 5. A positive linear trend was observed between age and AF, with older participants tending to have higher AF values; however, considerable scatter around the regression line reflected the dominant influence of treatment group. The relationship between age and PR was more striking: participants with `PR = yes` had a substantially higher median age of approximately 40 years compared to approximately 29 years for those without, with minimal overlap between the two interquartile ranges. This pattern strongly suggested that age was likely to confound the apparent treatment effect on PR, and motivated its inclusion as a covariate in all formal models.

Figure 6 displays the association between sex and both outcomes. AF distributions were similar between females and males, with median values of 60.2 and 63.5 respectively and substantially overlapping interquartile ranges. PR rates were virtually identical between the two sexes at 51.0% in both groups, confirming no meaningful association between sex and either outcome. Sex was nonetheless retained in formal modelling for completeness, though it was not expected to contribute meaningfully to either model.

Figure 7 summarises the exploratory SNP–outcome associations. Spearman correlations between SNP alternate allele dosage and AF were uniformly weak (largest: rs19486, $r = -0.231$). Heterogeneity of probability curves across dosage groups was noted for PR, though most SNPs showed only modest or inconsistent trends. No SNPs met the Bonferroni-corrected significance threshold of $p < 0.001$ for either outcome. This is not surprising given limited power across 50 independent tests with 100 study participants.

SNP Screening

To identify candidate genetic predictors for inclusion in the multivariable models, univariable regression screening was conducted separately for AF and PR across all 50 SNPs, excluding the zero-variance rs123456.

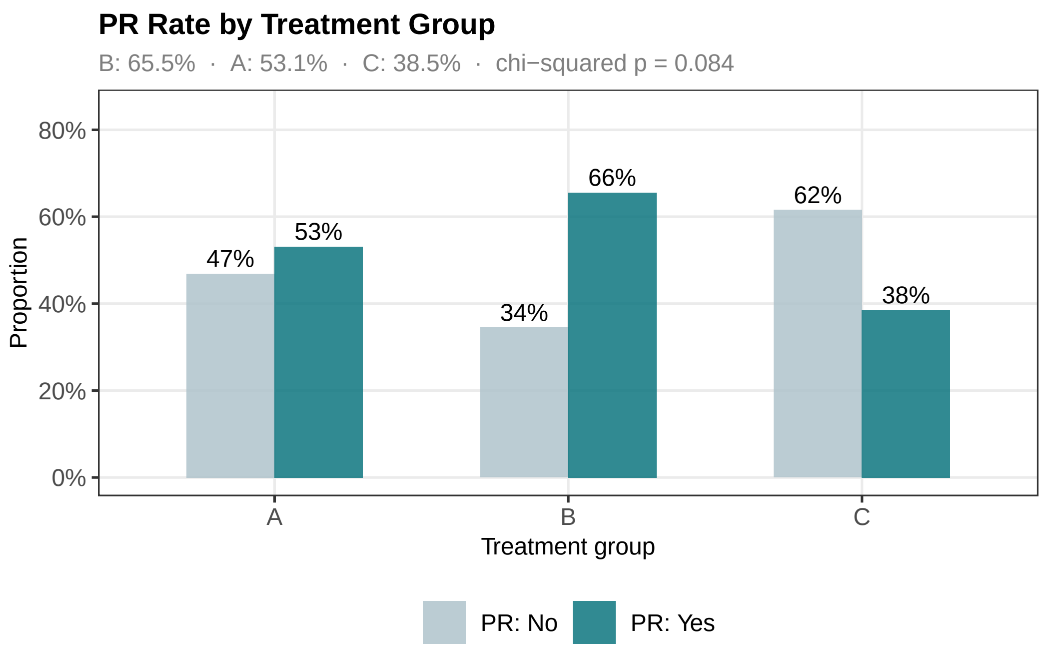

Figure 4: Unadjusted PR rate by treatment group for 100 participants. Bar height represents the proportion of participants with PR = yes or PR = no within each treatment group. Group B had the highest positive response rate (65.5%), followed by group A (53.1%) and group C (38.5%). The difference across groups did not reach statistical significance (chi-squared p = 0.084).

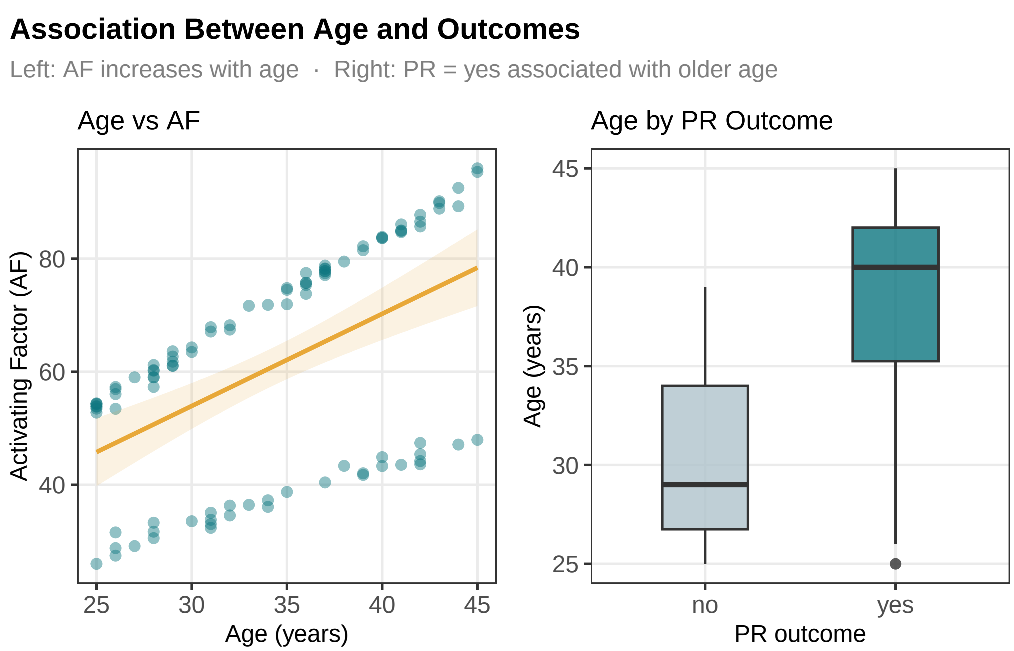

Figure 5: Association between age and the two outcomes of interest. Left: scatterplot of AF against age, showing a positive linear trend with the orange shaded region representing the 95% confidence band. Considerable scatter reflects the strong influence of treatment group on AF. Right: boxplot of age by PR outcome, showing that participants with PR = yes had a substantially higher median age (40 years) compared to those with PR = no (29 years), with minimal overlap between interquartile ranges.

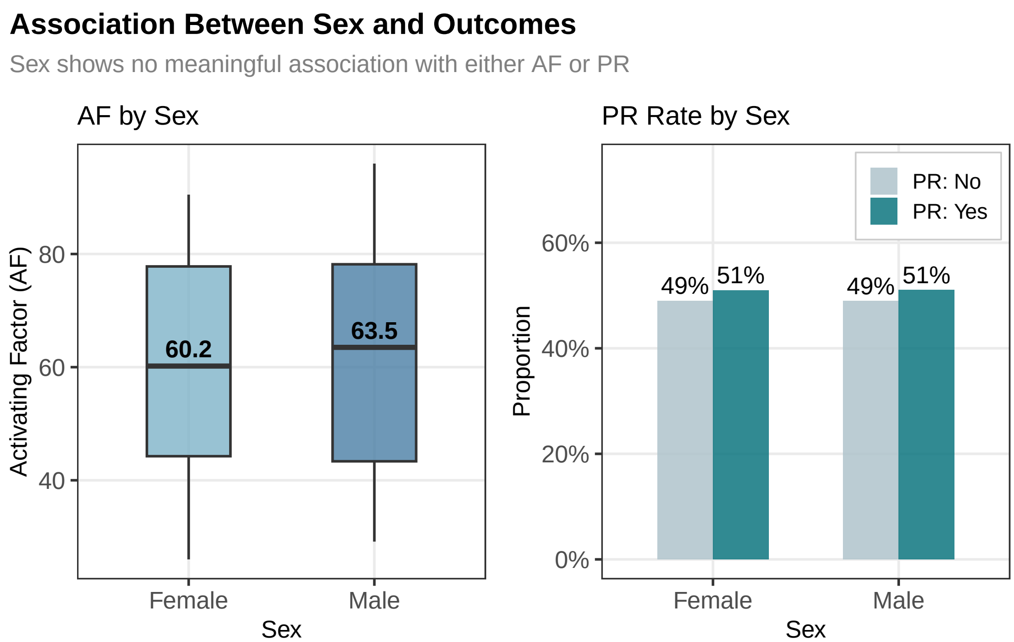

Figure 6: Association between sex and the two outcomes of interest. Left: boxplot of AF by sex, showing similar distributions between females (median = 60.2) and males (median = 63.5), with substantially overlapping interquartile ranges indicating no meaningful association between sex and AF. Right: PR rate by sex, showing virtually identical positive response rates in females (51.0%) and males (51.0%), indicating no association between sex and PR outcome.

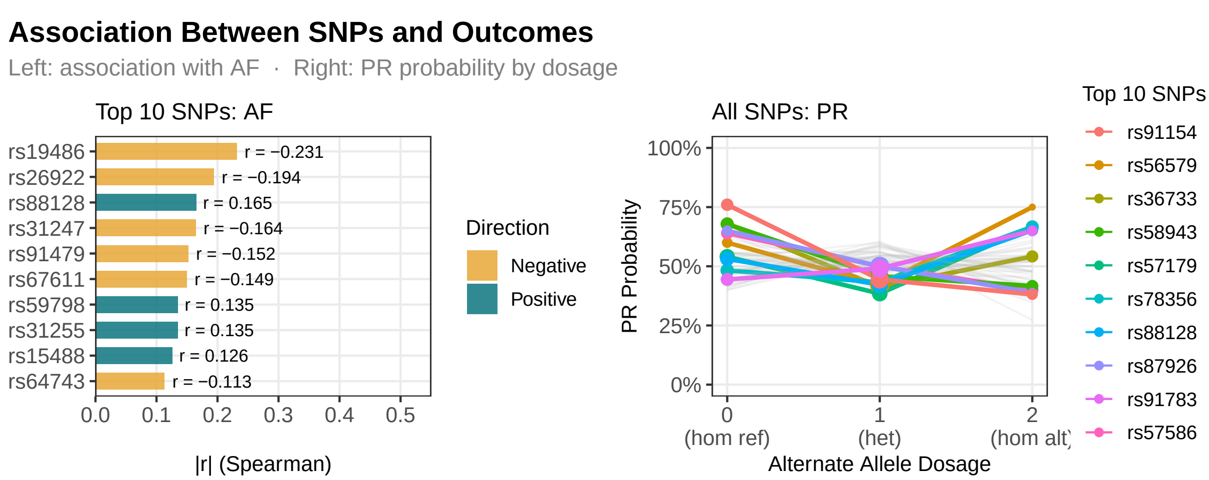

Figure 7: SNP screening results for both outcomes. Left: top 10 SNPs associated with AF ranked by Spearman correlation, with rs19486 showing the strongest negative association (r = -0.227). Right: PR probability by alternate allele dosage for all SNPs, with the top 10 by maximum PR probability highlighted. All correlations are weak and no SNPs survived Bonferroni correction; findings are exploratory only.

Linear regression was used for AF and logistic regression for PR. As anticipated given the small sample size, no SNPs survived the Bonferroni-corrected threshold for either outcome.

Applying the nominal threshold of $p < 0.05$ as a more lenient criterion, one SNP was retained per outcome. For AF, rs19486 passed the nominal threshold with $\beta = -6.23$ and $p = 0.015$, indicating that each additional alternate allele was associated with a decrease of 6.23 units in AF on average (Table 2). For PR, rs91154 passed the nominal threshold with OR = 0.438 and $p = 0.010$, suggesting that each additional alternate allele was associated with lower odds of a positive response (Table 3). Both findings were consistent with the Exploratory Data Analysis (EDA) and both SNPs were carried forward into the multivariable models as exploratory candidates, with all SNP-related findings treated as preliminary.

Table 2: Top 10 SNPs from univariable linear regression screening for AF, ranked by p-value.   represents the change in AF per additional alternate allele. No SNPs survived Bonferroni correction ($p < 0.001$); rs19486 passed the nominal threshold ($p < 0.05$) and was carried forward as a candidate predictor.

SNP	beta	p	Bonferroni (p<0.001)	Nominal (p<0.05)
rs19486	**-6.234**	**0.0149**	**No**	**Yes**
rs26922	-5.481	0.0529	No	No
rs31247	-4.634	0.0803	No	No
rs88128	4.258	0.1140	No	No
rs67611	-4.191	0.1301	No	No
rs91479	-3.856	0.1425	No	No
rs59798	3.267	0.2147	No	No
rs31255	3.695	0.2296	No	No
rs15488	3.237	0.2452	No	No
rs36733	-3.082	0.2782	No	No

Table 3: Top 10 SNPs from univariable logistic regression screening for PR, ranked by p-value. OR represents the change in odds of PR per additional alternate allele. No SNPs survived Bonferroni correction ($p < 0.001$); rs91154 passed the nominal threshold ($p < 0.05$) and was carried forward as a candidate predictor.

SNP	OR	p	Bonferroni (p<0.001)	Nominal (p<0.05)
rs91154	**0.438**	**0.0101**	**No**	**Yes**
rs58943	0.577	0.0562	No	No
rs57586	0.613	0.0929	No	No
rs29978	0.614	0.0949	No	No
rs87926	0.584	0.1101	No	No
rs69792	0.648	0.1134	No	No
rs13977	0.667	0.1616	No	No
rs59119	0.668	0.1627	No	No
rs35374	0.678	0.1696	No	No
rs78356	1.438	0.1819	No	No

AF Model

Three nested linear regression models were fitted for AF and compared using AIC and adjusted R^2 (Table 4), with formal nested model testing via ANOVA (Table 5). The treatment-only baseline model explained 70.5% of variance in AF, confirming that treatment group was a strong predictor on its own. Adding age and sex as covariates produced a dramatic improvement in model fit, with adjusted R^2 increasing to 0.977 and AIC decreasing by approximately 266 units (ANOVA $p < 0.001$), indicating that age in particular explained a substantial proportion of the residual variance left unexplained by treatment alone. The subsequent addition of rs19486 yielded no meaningful further improvement (ΔAIC < 1, ANOVA $p = 0.79$), suggesting that this SNP contributes negligible additional explanatory value once treatment and age are accounted for. The main-effects model comprising treatment, age, and sex was therefore selected as the final AF model.

Table 4: The table shows the comparison of three nested linear regression models for AF. The bolded row is the selected model. AIC is the Akaike Information Criterion where lower numbers denote a better fit. Adj. R^2 is the percent of variance in AF explained by the model.

Model	AIC	Adj. R^2
Treatment only	744.849	0.709
Main effects	**498.223**	**0.977**
Main + SNPs	499.792	0.977

Table 5: ANOVA nested model comparison for AF. Tests whether each additional set of predictors significantly improves model fit over the previous model. Res.Df: residual degrees of freedom; RSS: residual sum of squares; p-value: F-test for improvement.

term	df.residual	rss	df	sumsq	statistic	p.value
AF ~ treatment	95	10571.966	NA	NA	NA	NA
AF ~ treatment + age + sex	93	819.382	2	9752.584	549.920	0.000
AF ~ treatment + age + sex + as.numeric(rs19486)	92	815.789	1	3.593	0.405	0.526

The coefficient estimates from the selected model are presented in Table 6. After adjusting for age and sex, treatment B was associated with an AF value 35.7 units higher than treatment A (95% CI: 32.3 to 39.2, $p < 0.001$), and treatment C was associated with a value 34.3 units higher than A (95% CI: 31.0 to 37.7, $p < 0.001$). The magnitude and precision of these estimates confirm a clinically meaningful and statistically robust treatment effect that persists after covariate adjustment. Age was also strongly associated with AF, with each additional year corresponding to a 1.65-unit increase (95% CI: 1.53 to 1.77, $p < 0.001$). Sex had no meaningful effect on AF, with an estimated difference of 0.71 units that was far from statistical significance (95% CI: −0.56 to 1.97, $p = 0.27$). Overall, the model explained 97.7% of the variance in AF.

Table 6: Coefficient estimates from the selected AF linear regression model (treatment + age + sex, $n = 98$, adj. $R^2 = 0.977$). Treatment A is the reference group. beta: estimated change in AF per unit increase; SE: standard error; 95% CI: confidence interval; t: t-statistic; p: two-sided p-value.

Term	beta	SE	95% CI low	95% CI high	t	p
Treatment B vs A	35.722	0.779	34.175	37.269	45.853	0.0000
Treatment C vs A	34.339	0.719	32.912	35.767	47.770	0.0000
age	1.651	0.051	1.549	1.753	32.254	0.0000
Sex (Male vs Female)	0.709	0.616	-0.514	1.933	1.151	0.2526

PR Model

Three nested logistic regression models were fitted for PR and compared using AIC and McFadden's pseudo-R^2 (Table 7), with formal nested model testing via likelihood ratio tests (Table 8). The treatment-only baseline model had a McFadden R^2 of 0.030, suggesting that treatment group assigned at random explained close to none of the variation in PR outcome. Adding age and sex resulted in a substantial improvement in model fit (ΔMcFadden R^2 = 0.279, ΔAIC = -33.8; LRT $p < 0.001$), consistent with age explaining most of the PR variation. The addition of rs91154 to this model improved fit by only 1.3 AIC units (LRT p = 0.071) and was therefore not retained given the additional complexity of the model and failure of the SNP to pass Bonferroni correction. The main-effects model containing treatment group, age and sex was therefore selected as the final PR model.

Table 7: Table comparing three nested logistic regression models for PR. Boldface row indicates model chosen. AIC: Akaike Information Criterion; McFadden R^2: pseudo-R^2 measures improvement compared to a null intercept-only model.

Model	AIC	McFadden R^2
Treatment only	137.717	0.030
Main effects	**103.857**	**0.309**
Main + SNPs	102.593	0.346

Table 8: Likelihood ratio test for nested PR logistic regression models. Tests whether adding each additional set of predictors significantly improves model fit. Df: degrees of freedom added; Deviance: decrease in residual deviance; p-value: chi-squared test of improvement.

term	df.residual	residual.deviance	df	deviance	p.value
PR ~ treatment	95	131.717	NA	NA	NA
PR ~ treatment + age + sex	93	93.857	2	37.860	0.000
PR ~ treatment + age + sex + as.numeric(rs91154)	92	90.593	1	3.264	0.071

Odds ratios estimated from the selected model are shown in Table 9. Age was by far the strongest predictor of PR. For every additional year of age, there was a 1.30 times greater odds of having a positive response (95% CI: 1.18 to 1.46, $p < 0.001$). Neither treatment B nor treatment C had significantly different odds of PR compared to treatment A after adjusting for age and sex – the estimated odds ratios were 0.975 (95% CI: 0.27 to 3.53, p = 0.97) and 0.715 (95% CI: 0.21 to 2.37, p = 0.58) respectively. This is in contrast to the unadjusted positive response rates we observed in EDA, where it seemed like group B had a substantially higher rate of positive response compared to group C. The attenuation of this effect upon adjustment for age strongly suggests that the apparent treatment effect on PR was driven by confounding from age rather than a true treatment effect. Sex was not meaningfully associated with PR (OR = 0.49, 95% CI: 0.16 to 1.37, p = 0.19).

Table 9: Odds Ratios from Fitted PR Logistic Regression Model (Treatment + Age + Sex, n = 98). Treatment A is reference group. Bold row indicates dominant predictor. OR = odds ratio (exponentiated log-odds coefficient); 95% CI = confidence interval; p = two-sided Wald test p-value.

Term	OR	95% CI low	95% CI high	p
Treatment B vs A	0.975	0.266	3.534	0.9695
Treatment C vs A	0.715	0.213	2.370	0.5804
age	**1.303**	**1.184**	**1.460**	**0.0000**
Sex (Male vs Female)	0.488	0.160	1.374	0.1869

Assumption Checking

Diagnostic checks were conducted for both the AF linear regression model and the PR logistic regression model to assess whether key modelling assumptions were satisfied.

The four residual diagnostic plots for the AF model are presented in Figure 8. There was a slight U-shaped trend in residuals vs fitted, indicating a slight departure from linearity, and variance wasn't completely constant over the range of fitted values. The points were reasonably close to the line in the Q-Q plot, except for slight deviations in the tails. So, normality was roughly met. Cook's distance flagged three potentially influential observations (IDs 53, 78, and 85). They should be checked for data entry mistakes, but should not be removed unless there is a good reason to do so. The VIFs for all predictors were near 1 (Table 10), indicating that there wasn't any substantial multicollinearity between treatment, age, and sex.

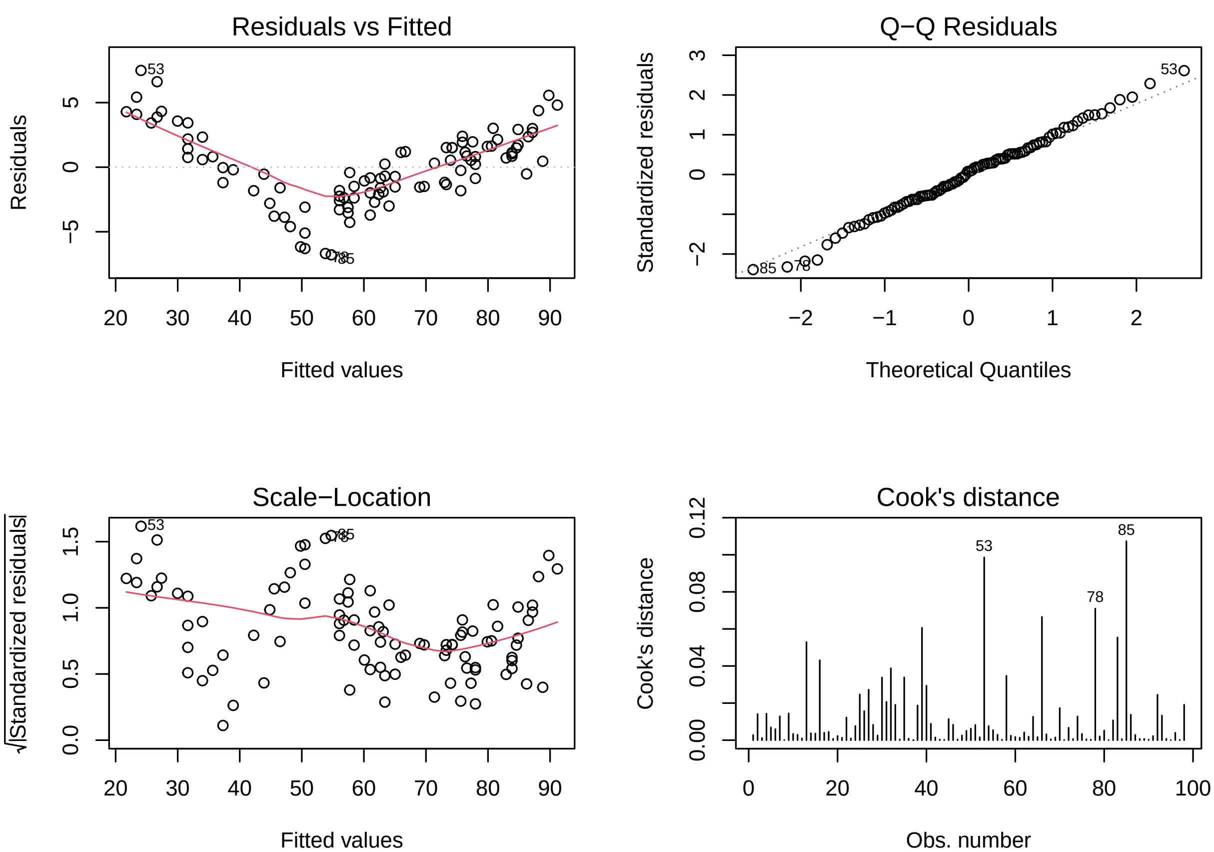

Figure 8: Diagnostic plots from linear regression of the chosen AF model (treatment + age + sex, n = 98). The residuals versus fitted plot displays mild U-shape indicating slight non-linearity; the Q-Q plot is reasonable aside from some deviations in the tails; Cook's distance identifies observations 53, 78, and 85 as possibly influential.

Table 10: Variance inflation factors (VIF) associated with the final AF linear regression model. GVIF^(1/2Df) values near 1 suggest that there is no substantial multicollinearity between predictors. Values >2.5 would indicate potential concern.

Predictor	GVIF^(1/2Df)
treatment	1.022

age	1.058
sex	1.027

Diagnostic results for the PR logistic regression model are summarised in Table 11 and Figure 9. The Box-Tidwell procedure produced a marginally significant coefficient for the interaction term with age (β = 1.48, p = 0.046). There may therefore be some non-linearity between age and log odds of PR; however with the small sample it is more interested in this not violating the assumption than in the result itself. Tabulations of treatment and sex against PR revealed there were no zero cells (minimum cell n = 10), so complete separation was ruled out, as was quasi-complete separation. The Hosmer-Lemeshow test was not significant ($\chi^2 = 8.61$, df = 8, p = 0.376), suggesting the model did not do a significantly poor job of predicting observed outcome frequencies within deciles of predicted probability. VIF were near 1 for all predictor terms, so multicollinearity was not present. Seven observations exceeded the Cook's distance threshold of 4/n (participants 16, 40, 49, 57, 72, 82, and 92), with participants 49 and 82 showing the largest values; as with the AF model, these observations warrant inspection but removal is not recommended without substantive justification.

Table 11: Diagnostics for the chosen PR logistic regression model (treatment + age + sex; n=98). Box-Tidwell tests linearity of age with respect to log-odds of PR; Hosmer-Lemeshow tests overall fit across deciles of predicted probability; VIF tests for multicollinearity.

Check	Result	Conclusion
Box-Tidwell (age linearity)	beta = 1.478, p = 0.046	Borderline; interpret with caution
Hosmer-Lemeshow goodness-of-fit	chi-squared = 8.61, df = 8, p = 0.376	Adequate fit
Complete separation	No (min cell n = 10)	No separation
VIF: treatment	1.008	No multicollinearity
VIF: age	1.071	No multicollinearity
VIF: sex	1.059	No multicollinearity

Discussion

This analysis tested if treatment group significantly affected AF and PR after adjusting for possible confounders age, sex, and genotype. The analysis found significant evidence of treatment affecting AF, but not conclusively for PR.

Treatment group was a strong and statistically robust predictor of AF. Both B and C groups had markedly higher AF values than group A – by ~35.7 and 34.3 units respectively – and this effect remained after accounting for age and sex: the final model accounted for 97.7% of variance in AF. Age was also independently associated with AF: For every year older a participant was, their AF value went up by 1.65 units. Both treatment assignment and participant age appear to have useful effects on AF. Regression slopes did not significantly differ across treatment groups in the EDA, indicating that the effect of age was parallel across treatments (i.e. there was no interaction). Sex was not meaningfully involved in the model; this makes sense given that there was no marginal association with sex in the EDA.

The results for PR were more nuanced. While treatment group B seems to clearly have a higher PR rate than C in the unadjusted analysis (65.5% vs 38.5%), this did not reach conventional statistical significance (chi-squared p = 0.084). And when controlling for age in the logistic regression model, the difference between B and C decreased substantially - neither B nor C were significantly different from A in the odds of PR. Age was by far the strongest predictor of PR - every additional year of age was associated with 1.30 times greater odds of PR. The unadjusted differences between treatment groups in PR from the EDA were therefore mostly due to confounding by age: members of group B just so happened to be older, on average, and older individuals were more likely to respond positively regardless of treatment assignment. When age was controlled for, treatment group was no longer associated with a statistically significant amount of the variability in PR.

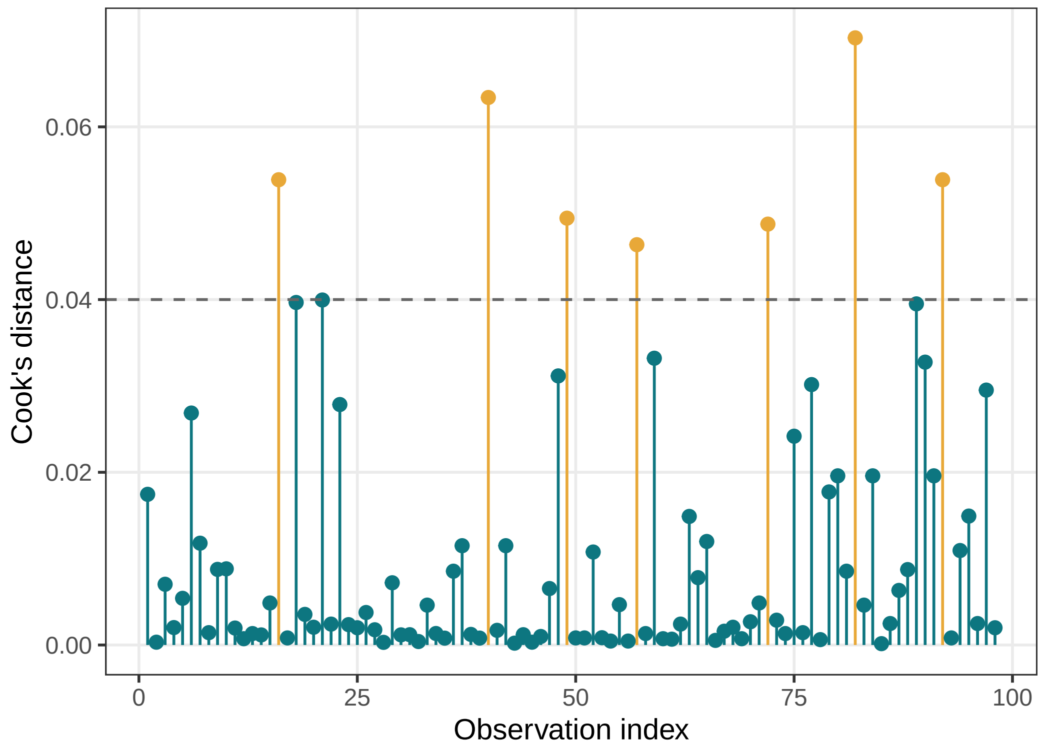

Figure 9: Cook's distance of the PR logistic model (model contains treatment + age + sex, n = 98). Observations flagged with orange are larger than 4/n. Seven points are flagged (IDs: 16, 40, 49, 57, 72, 82, 92). IDs 49 and 82 have the highest values.

These genetic exploratory analyses were hypothesis-generating and should be interpreted very cautiously. No SNPs remained significant after Bonferroni correction for either outcome; this was not surprising given the modest statistical power we had with 100 individuals for 50 tests performed simultaneously. The two nominally-significant candidates (rs19486 for AF, rs91154 for PR) explained trivial additional amounts of variation in these multivariable models after treatment and age, and neither result should be believed until replicated in an independent sample.

SSeveral caveats to these findings are worth mentioning. Firstly, our sample size of n = 100 participants is quite small, affording us limited power. This is particularly true for the genetic analyses, and also for detecting small-to-moderate treatment effects on PR. The fact that treatment did not achieve significance for PR (p = 0.084, unadjusted) may be due to a true lack of effect, but could alternatively be due to insufficient power to detect a small to moderate sized effect. Secondly, our borderline Box-Tidwell result for the PR model (p = 0.046) indicates that there may be some non-linear relationship between age and log-odds of PR that is not being modelled by our logistic regression model. Including a more flexible terms for age in the model (e.g. a quadratic age term, or splines) would be worthwhile in future analyses. Thirdly, our treatment groups were not well balanced for age, which is why we needed to adjust for it as a covariate. But this result also begs the question of whether our randomisation process was truly adequate. To the extent that age differs across groups due to systematic (non-random) allocation, residual confounding may remain after covariate adjustment. Finally, the SNP screening methodology employed here has an inflated false positive rate at the nominal alpha level, so any SNPs identified here should be considered hypothesis-generating rather than confirmatory.

A larger sample size in future studies would permit more robust SNP screening using Bonferroni-corrected thresholds, increased power to examine treatment effects on PR and allow more flexible modelling of the age–PR relationship. Validation of the rs19486 and rs91154 associations in an independent cohort would also provide insight into whether these are true positive findings or false positives.

Appendix

```r
# PACKAGES
pacman::p_load(tidyverse, readxl, table1, patchwork, scales, broom, ResourceSelection,
               car, kableExtra)

# COLORS
trt_colours <- c(A = "#0D7680", B = "#E8A838", C = "#A06030")

# READ IN DATA
df_fitfr_raw <- read_excel("FITFR-final-data-v3.xlsx", sheet = 1, skip = 1)

# REMOVE JUNK ROWS
df_fitfr <- df_fitfr_raw |>
  filter(!is.na(ID))

# CLEAN & RECODE VARIABLES
df_fitfr <- df_fitfr |>
  mutate(
    ID        = as.integer(ID),
    age       = as.numeric(age) |> na_if(-99),
    AF        = as.numeric(AF),
    sex       = factor(case_when(
                  str_to_lower(sex) == "female"         ~ "Female",
                  str_to_lower(sex) %in% c("male", "m") ~ "Male"),
                  levels = c("Female", "Male")),
    treatment = factor(str_to_upper(treatment), levels = c("A", "B", "C")),
```

```r
    smoker    = factor(smoker, levels = "N"),
    PR        = factor(case_when(
                  str_to_lower(PR) == "yes" ~ 1L,
                  str_to_lower(PR) == "no"  ~ 0L),
                  levels = c(0, 1), labels = c("no", "yes"))
  )

# SNP ENCODING
snp_cols <- names(df_fitfr)[str_detect(names(df_fitfr), "^rs")]

encode_snp <- function(x) {
  x       <- as.character(x)
  genos   <- na.omit(unique(x))
  homozyg <- genos[substr(genos, 1, 1) == substr(genos, 2, 2)]
  if (length(homozyg) < 1) return(factor(x))
  counts      <- table(x[x %in% homozyg])
  ref_allele  <- substr(names(which.max(counts)), 1, 1)
  alt_alleles <- setdiff(unique(unlist(strsplit(genos, ""))), ref_allele)
  count_alt   <- function(g) sum(strsplit(g, "")[[1]] %in% alt_alleles)
  coded       <- sapply(x, function(g) if (is.na(g)) NA_integer_ else count_alt(g))
  factor(coded, levels = c(0, 1, 2), ordered = TRUE)
}

df_fitfr[snp_cols] <- lapply(df_fitfr[snp_cols], encode_snp)

# TABLE 1
table1(
  ~ age + sex + AF + PR | treatment,
  data = df_fitfr
)

# DISTRIBUTION PLOTS: CONTINUOUS VARIABLES
p_af <- ggplot(df_fitfr, aes(x = AF)) +
  geom_histogram(bins = 30, fill = "#0D7680", alpha = 0.8, colour = "white") +
  geom_density(aes(y = after_stat(count)), colour = "gray30", linewidth = 0.7) +
  labs(title = "Activating Factor (AF)", x = "AF", y = "Count") +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank()
  )

p_age <- ggplot(df_fitfr |> filter(!is.na(age)), aes(x = age)) +
  geom_histogram(bins = 20, fill = "#A06030", alpha = 0.8, colour = "white") +
  geom_density(aes(y = after_stat(count)), colour = "gray30", linewidth = 0.7) +
  labs(title = "Age", x = "Age (years)", y = "Count") +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
```

```r
    panel.grid.minor = element_blank()
  )

p_af + p_age +
  plot_annotation(
    title    = "Distribution of Continuous Variables",
    subtitle = "Dark Gary curve = density estimate",
    theme    = theme(
      plot.title    = element_text(hjust = 0, face = "bold", size = 11),
      plot.subtitle = element_text(hjust = 0, size = 9, colour = "grey50")
    )
  )

# DISTRIBUTION PLOTS: CATEGORICAL VARIABLES
p_treat <- ggplot(df_fitfr, aes(x = treatment, fill = treatment)) +
  geom_bar(alpha = 0.85) +
  geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.5, size = 3) +
  scale_fill_manual(values = trt_colours) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +
  labs(title = "Treatment", x = "Treatment group", y = "Count") +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank(),
    legend.position  = "none"
  )

p_sex <- ggplot(df_fitfr, aes(x = sex, fill = sex)) +
  geom_bar(alpha = 0.85) +
  geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.5, size = 3) +
  scale_fill_manual(values = c(Female = "#7FB3C8", Male = "#4A7FA5")) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +
  labs(title = "Sex", x = "Sex", y = "Count") +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank(),
    legend.position  = "none"
  )

p_pr <- ggplot(df_fitfr, aes(x = PR, fill = PR)) +
  geom_bar(alpha = 0.85) +
  geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.5, size = 3) +
  scale_fill_manual(values = c(no = "#B0C4CC", yes = "#0D7680")) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +
  labs(title = "PR Outcome", x = "PR", y = "Count") +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10),
```

```
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank(),
    legend.position  = "none"
  )

p_treat + p_sex + p_pr +
  plot_annotation(
    title = "Distribution of Categorical Variables",
    subtitle = "Numbers above bars indicate participant counts",
    theme = theme(
      plot.title = element_text(hjust = 0, face = "bold", size = 11),
      plot.subtitle = element_text(hjust = 0, size = 9, colour = "grey50")
    )
  )

# PRIMARY RELATIONSHIPS: TREATMENT VS OUTCOMES
p_treat_af <- ggplot(df_fitfr, aes(x = treatment, y = AF, fill = treatment)) +
  geom_boxplot(alpha = 0.8, outlier.shape = 21, width = 0.5) +
  stat_summary(fun = mean, geom = "point", shape = 23,
               size = 3, fill = "white", colour = "black") +
  scale_fill_manual(values = trt_colours) +
  labs(
    title    = "AF by Treatment Group",
    subtitle = "Diamond = group mean  ·  Kruskal-Wallis p < 0.001",
    x        = "Treatment group",
    y        = "Activating Factor (AF)"
  ) +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(face = "bold", size = 11, hjust = 0),
    plot.subtitle    = element_text(size = 9, colour = "grey50", hjust = 0),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank(),
    legend.position  = "none"
  )
p_treat_af

p_treat_pr <- df_fitfr |>
  filter(!is.na(PR)) |>
  group_by(treatment, PR) |>
  summarise(n = n(), .groups = "drop") |>
  mutate(prop = n / sum(n), .by = treatment) |>
  ggplot(aes(x = treatment, y = prop, fill = PR)) +
  geom_col(position = "dodge", alpha = 0.85, width = 0.6) +
  geom_text(aes(label = percent(prop, accuracy = 1)),
            position = position_dodge(0.6), vjust = -0.4, size = 3) +
  scale_fill_manual(values = c(no = "#B0C4CC", yes = "#0D7680"),
                    labels = c("PR: No", "PR: Yes")) +
  scale_y_continuous(labels = percent, limits = c(0, 0.85)) +
  labs(
    title    = "PR Rate by Treatment Group",
```

```
    subtitle = "B: 65.5%  ·  A: 53.1%  ·  C: 38.5%  ·  chi-squared p = 0.084",
    x        = "Treatment group",
    y        = "Proportion",
    fill     = NULL
  ) +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(face = "bold", size = 11, hjust = 0),
    plot.subtitle    = element_text(size = 9, colour = "grey50", hjust = 0),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank(),
    legend.position  = "bottom",
    legend.text      = element_text(size = 9)
  )
p_treat_pr

# PRIMARY RELATIONSHIPS: AGE VS OUTCOMES
p_age_af <- ggplot(df_fitfr, aes(x = age, y = AF)) +
  geom_point(alpha = 0.45, colour = "#0D7680", size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, colour = "#E8A838",
              fill = "#E8A838", alpha = 0.15, linewidth = 0.8) +
  labs(
    title = "Age vs AF",
    x     = "Age (years)",
    y     = "Activating Factor (AF)"
  ) +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10, hjust = 0),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank()
  )

p_age_pr <- ggplot(df_fitfr |> filter(!is.na(PR)), aes(x = PR, y = age, fill = PR)) +
  geom_boxplot(alpha = 0.8, width = 0.45) +
  scale_fill_manual(values = c(no = "#B0C4CC", yes = "#0D7680")) +
  labs(
    title = "Age by PR Outcome",
    x     = "PR outcome",
    y     = "Age (years)"
  ) +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10, hjust = 0),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank(),
    legend.position  = "none"
  )

p_age_af + p_age_pr +
```

```r
  plot_annotation(
    title = "Association Between Age and Outcomes",
    subtitle = "Left: AF increases with age  ·  Right: PR = yes associated with older age",
    theme = theme(
      plot.title    = element_text(hjust = 0, face = "bold", size = 11),
      plot.subtitle = element_text(hjust = 0, size = 9, colour = "grey50")
    )
  )

# PRIMARY RELATIONSHIPS: SEX VS OUTCOMES
p_sex_af <- ggplot(df_fitfr, aes(x = sex, y = AF, fill = sex)) +
  geom_boxplot(alpha = 0.8, width = 0.45) +
  stat_summary(
    fun      = median,
    geom     = "text",
    aes(label = round(after_stat(y), 1)),
    vjust    = -0.5,
    size     = 3,
    fontface = "bold"
  ) +
  scale_fill_manual(values = c(Female = "#7FB3C8", Male = "#4A7FA5")) +
  labs(
    title = "AF by Sex",
    x     = "Sex",
    y     = "Activating Factor (AF)"
  ) +
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10, hjust = 0),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank(),
    legend.position  = "none"
  )

p_sex_pr <- df_fitfr |>
  filter(!is.na(PR), !is.na(sex)) |>
  group_by(sex, PR) |>
  summarise(n = n(), .groups = "drop") |>
  mutate(prop = n / sum(n), .by = sex) |>
  ggplot(aes(x = sex, y = prop, fill = PR)) +
  geom_col(position = "dodge", alpha = 0.85, width = 0.6) +
  geom_text(aes(label = percent(prop, accuracy = 1)),
            position = position_dodge(0.6), vjust = -0.4, size = 3) +
  scale_fill_manual(values = c(no = "#B0C4CC", yes = "#0D7680"),
                    labels = c("PR: No", "PR: Yes")) +
  scale_y_continuous(labels = percent, limits = c(0, 0.75)) +
  labs(
    title = "PR Rate by Sex",
    x     = "Sex",
    y     = "Proportion",
    fill  = NULL
  ) +
```

```r
  theme_bw(base_size = 10) +
  theme(
    plot.title       = element_text(size = 10, hjust = 0),
    axis.text        = element_text(size = 9),
    axis.title       = element_text(size = 9),
    panel.grid.minor = element_blank(),
    legend.position  = c(0.8, 0.88),
    legend.background = element_rect(fill = "white", colour = "grey80", linewidth = 0.3),
    legend.text      = element_text(size = 8),
    legend.key.size  = unit(0.4, "cm")
  )

p_sex_af + p_sex_pr +
  plot_annotation(
    title    = "Association Between Sex and Outcomes",
    subtitle = "Sex shows no meaningful association with either AF or PR",
    theme    = theme(
      plot.title    = element_text(hjust = 0, face = "bold", size = 11),
      plot.subtitle = element_text(hjust = 0, size = 9, colour = "grey50")
    )
  )

# PRIMARY RELATIONSHIPS: SNPS VS OUTCOMES
snp_cols_clean <- names(df_fitfr)[str_detect(names(df_fitfr), "^rs") &
                                  names(df_fitfr) != "rs123456"]

snp_cor_af <- tibble(
  SNP = snp_cols_clean,
  r   = sapply(snp_cols_clean, function(s) {
    cor(as.numeric(df_fitfr[[s]]), df_fitfr$AF, use = "complete.obs", method = "spearman")
  })
) |>
  arrange(desc(abs(r))) |>
  slice_head(n = 10) |>
  mutate(
    SNP       = fct_reorder(SNP, abs(r)),
    direction = if_else(r > 0, "Positive", "Negative")
  )

p_snp_af <- ggplot(snp_cor_af, aes(x = abs(r), y = SNP, fill = direction)) +
  geom_col(width = 0.65, alpha = 0.85) +
  geom_text(aes(label = sprintf("r = %.3f", r)),
            hjust = -0.1, size = 3) +
  scale_fill_manual(values = c(Positive = "#0D7680", Negative = "#E8A838")) +
  scale_x_continuous(limits = c(0, 0.55), expand = c(0, 0)) +
  labs(
    title = "Top 10 SNPs: AF",
    x     = "|r| (Spearman)",
    y     = NULL,
    fill  = "Direction"
  ) +
  theme_bw(base_size = 12) +
  theme(
```

```r
    plot.title       = element_text(size = 12, hjust = 0),
    axis.text        = element_text(size = 11),
    axis.title       = element_text(size = 11),
    panel.grid.minor = element_blank(),
    legend.position  = "right",
    legend.text      = element_text(size = 10),
    legend.title     = element_text(size = 11)
  )

snp_probs <- df_fitfr |>
  select(PR, all_of(snp_cols_clean)) |>
  pivot_longer(-PR, names_to = "SNP", values_to = "dosage") |>
  filter(!is.na(dosage), !is.na(PR)) |>
  group_by(SNP, dosage) |>
  summarise(prob = mean(PR == "yes"), n = n(), .groups = "drop") |>
  mutate(dosage = as.numeric(as.character(dosage)))

top10_snps <- snp_probs |>
  group_by(SNP) |>
  summarise(max_prob = max(prob)) |>
  arrange(desc(max_prob)) |>
  slice_head(n = 10) |>
  pull(SNP)

p_snp_pr <- snp_probs |>
  mutate(
    highlight = if_else(SNP %in% top10_snps, SNP, "other"),
    alpha_val = if_else(SNP %in% top10_snps, 1, 0.15),
    lwd_val   = if_else(SNP %in% top10_snps, 1.1, 0.5)
  ) |>
  ggplot(aes(x = dosage, y = prob, group = SNP,
             colour = highlight, alpha = alpha_val, linewidth = lwd_val)) +
  geom_line() +
  geom_point(data = ~ filter(., SNP %in% top10_snps),
             aes(size = n), alpha = 1) +
  scale_colour_manual(
    values = c(setNames(scales::hue_pal()(10), top10_snps), other = "grey70"),
    breaks = top10_snps
  ) +
  scale_alpha_identity() +
  scale_linewidth_identity() +
  scale_size_continuous(range = c(1, 4), guide = "none") +
  scale_y_continuous(labels = percent, limits = c(0, 1)) +
  scale_x_continuous(breaks = c(0, 1, 2),
                     labels = c("0\n(hom ref)", "1\n(het)", "2\n(hom alt)")) +
  labs(
    title  = "All SNPs: PR",
    x      = "Alternate Allele Dosage",
    y      = "PR Probability",
    colour = "Top 10 SNPs"
  ) +
  theme_bw(base_size = 12) +
  theme(
```

```r
    plot.title       = element_text(size = 12, hjust = 0),
    axis.text        = element_text(size = 11),
    axis.title       = element_text(size = 11),
    panel.grid.minor = element_blank(),
    legend.position  = "right",
    legend.text      = element_text(size = 10),
    legend.title     = element_text(size = 11)
  )

p_snp_af + p_snp_pr +
  plot_annotation(
    title    = "Association Between SNPs and Outcomes",
    subtitle = "Left: association with AF  ·  Right: PR probability by dosage",
    theme    = theme(
      plot.title    = element_text(hjust = 0, face = "bold", size = 15),
      plot.subtitle = element_text(hjust = 0, size = 12, colour = "grey50")
    )
  )

# SCREENING SNP FOR AF MODEL
snp_cols_model <- names(df_fitfr)[str_detect(names(df_fitfr), "^rs")
                                  & names(df_fitfr) != "rs123456"]


snp_screen_af <- tibble(
  SNP  = snp_cols_model,
  p    = sapply(snp_cols_model, function(s) {
    summary(lm(AF ~ as.numeric(df_fitfr[[s]]), data = df_fitfr))$coefficients[2, 4]
  }),
  beta = sapply(snp_cols_model, function(s) {
    coef(lm(AF ~ as.numeric(df_fitfr[[s]]), data = df_fitfr))[2]
  })
) |>
  arrange(p) |>
  mutate(bonferroni = p < 0.05 / n(),          # Bonferroni-corrected threshold
         nominal    = p < 0.05)

cat("SNPs passing Bonferroni threshold (AF):", sum(snp_screen_af$bonferroni), "\n")
cat("SNPs passing nominal p < 0.05 (AF):",    sum(snp_screen_af$nominal),    "\n")
print(head(snp_screen_af, 10))
snp_screen_af |>
  slice_head(n = 10) |>
  mutate(
    beta = round(beta, 3),
    p    = formatC(p, format = "f", digits = 4),
    bonferroni = if_else(bonferroni, "Yes", "No"),
    nominal    = if_else(nominal,    "Yes", "No")
  ) |>
  select(SNP, beta, p, `Bonferroni (p<0.001)` = bonferroni,
         `Nominal (p<0.05)` = nominal) |>
  kable(align = "lrrrr",
        caption = "Top 10 SNPs from univariable linear regression screening for AF,
        ranked by p-value.  represents the change in AF per additional alternate allele.
```

```r
        No SNPs survived Bonferroni correction (p < 0.001); rs19486 passed the nominal
        threshold (p < 0.05) and was carried forward as a candidate predictor.") |>
  kable_styling(bootstrap_options = c("striped", "condensed"),
                full_width = FALSE) |>
  row_spec(which(snp_screen_af$nominal[1:10]), bold = TRUE)

# SCREENING SNP FOR PR MODEL
pr_num <- as.numeric(df_fitfr$PR) - 1   # recode to 0/1

snp_screen_pr <- tibble(
  SNP  = snp_cols_model,
  p    = sapply(snp_cols_model, function(s) {
    summary(glm(pr_num ~ as.numeric(df_fitfr[[s]]),
                family = binomial))$coefficients[2, 4]
  }),
  OR   = sapply(snp_cols_model, function(s) {
    exp(coef(glm(pr_num ~ as.numeric(df_fitfr[[s]]),
                 family = binomial))[2])
  })
) |>
  arrange(p) |>
  mutate(bonferroni = p < 0.05 / n(),
         nominal    = p < 0.05)

cat("\nSNPs passing Bonferroni threshold (PR):", sum(snp_screen_pr$bonferroni), "\n")
cat("SNPs passing nominal p < 0.05 (PR):",    sum(snp_screen_pr$nominal),    "\n")
print(head(snp_screen_pr, 10))

top_snps_af <- snp_screen_af |> filter(nominal) |> pull(SNP)
top_snps_pr <- snp_screen_pr |> filter(nominal) |> pull(SNP)

cat("Top AF SNP candidates:", paste(top_snps_af, collapse = ", "), "\n")
cat("Top PR SNP candidates:", paste(top_snps_pr, collapse = ", "), "\n")

snp_screen_pr |>
  slice_head(n = 10) |>
  mutate(
    OR   = round(OR, 3),
    p    = formatC(p, format = "f", digits = 4),
    bonferroni = if_else(bonferroni, "Yes", "No"),
    nominal    = if_else(nominal,    "Yes", "No")
  ) |>
  select(SNP, OR, p, `Bonferroni (p<0.001)` = bonferroni, `Nominal (p<0.05)` = nominal) |>
  kable(align = "lrrrr",
        caption = "Top 10 SNPs from univariable logistic regression screening for PR,
        ranked by p-value. OR represents the change in odds of PR per additional alternate
        allele. No SNPs survived Bonferroni correction (p < 0.001); rs91154 passed the
        nominal threshold (p < 0.05) and was carried forward as a candidate predictor.") |>
  kable_styling(bootstrap_options = c("striped", "condensed"),
                full_width = FALSE) |>
  row_spec(which(snp_screen_pr$nominal[1:10]), bold = TRUE)

# AF MODEL
```

```r
df_model <- df_fitfr |> drop_na(age)

lm_af_base <- lm(AF ~ treatment,             data = df_model)
lm_af_main <- lm(AF ~ treatment + age + sex, data = df_model)

if (length(top_snps_af) > 0) {
  snp_terms_af <- paste(sprintf("as.numeric(%s)", top_snps_af), collapse = " + ")
  lm_af_snp <- lm(as.formula(paste("AF ~ treatment + age + sex +", snp_terms_af)),
                  data = df_fitfr)
} else {
  lm_af_snp <- lm_af_main
}

tibble(
  Model    = c("Treatment only", "Main effects", "Main + SNPs"),
  AIC      = c(AIC(lm_af_base), AIC(lm_af_main), AIC(lm_af_snp)),
  `Adj. R²` = c(
    summary(lm_af_base)$adj.r.squared,
    summary(lm_af_main)$adj.r.squared,
    summary(lm_af_snp)$adj.r.squared
  )
) |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  kable(caption = "The table shows the comparison of three nested linear regression
        models for AF. The bolded row is the selected model. AIC is the Akaike
        Information Criterion where lower numbers denote a better fit. Adj. R²
        is the percent of variance in AF explained by the model.") |>
  kable_styling(bootstrap_options = c("striped", "condensed"), full_width = FALSE) |>
  row_spec(2, bold = TRUE)

anova(lm_af_base, lm_af_main, lm_af_snp) |>
  tidy() |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  kable(caption = "ANOVA nested model comparison for AF. Tests whether each additional
        set of predictors significantly improves model fit over the previous model.
        Res.Df: residual degrees of freedom; RSS: residual sum of squares; p-value:
        F-test for improvement.") |>
  kable_styling(bootstrap_options = c("striped", "condensed"), full_width = FALSE)

tidy(lm_af_main, conf.int = TRUE) |>
  filter(term != "(Intercept)") |>
  mutate(
    term = case_when(
      term == "treatmentB" ~ "Treatment B vs A",
      term == "treatmentC" ~ "Treatment C vs A",
      term == "sexMale"    ~ "Sex (Male vs Female)",
      .default = term
    ),
    across(c(estimate, conf.low, conf.high, std.error, statistic), ~ round(.x, 3)),
    p.value = formatC(p.value, format = "f", digits = 4)
  ) |>
  select(Term = term, beta = estimate, SE = std.error,
         `95% CI low` = conf.low, `95% CI high` = conf.high,
```

```
         t = statistic, p = p.value) |>
  kable(caption = "Coefficient estimates from the selected AF linear regression model
        (treatment + age + sex, n = 98, adj. R² = 0.977). Treatment A is the reference
        group. beta: estimated change in AF per unit increase; SE: standard error;
        95% CI: confidence interval; t: t-statistic; p: two-sided p-value.") |>
  kable_styling(bootstrap_options = c("striped", "condensed"), full_width = FALSE)

# PR MODEL
glm_pr_base <- glm(PR ~ treatment,             data = df_model, family = binomial)
glm_pr_main <- glm(PR ~ treatment + age + sex, data = df_model, family = binomial)

if (length(top_snps_pr) > 0) {
  snp_terms_pr <- paste(sprintf("as.numeric(%s)", top_snps_pr), collapse = " + ")
  glm_pr_snp <- glm(as.formula(paste("PR ~ treatment + age + sex +", snp_terms_pr)),
                    data = df_fitfr, family = binomial)
} else {
  glm_pr_snp <- glm_pr_main
}

mcfadden_r2 <- function(model) {
  1 - (logLik(model) / logLik(update(model, . ~ 1)))
}

tibble(
  Model        = c("Treatment only", "Main effects", "Main + SNPs"),
  AIC          = c(AIC(glm_pr_base), AIC(glm_pr_main), AIC(glm_pr_snp)),
  `McFadden R²` = c(
    as.numeric(mcfadden_r2(glm_pr_base)),
    as.numeric(mcfadden_r2(glm_pr_main)),
    as.numeric(mcfadden_r2(glm_pr_snp))
  )
) |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  kable(caption = "Table comparing three nested logistic regression models for PR.
        Boldface row indicates model chosen. AIC: Akaike Information Criterion; McFadden
        R²: pseudo-R² measures improvement compared to a null intercept-only model.") |>
  kable_styling(bootstrap_options = c("striped", "condensed"), full_width = FALSE) |>
  row_spec(2, bold = TRUE)

anova(glm_pr_base, glm_pr_main, glm_pr_snp, test = "LRT") |>
  tidy() |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  kable(caption = "Likelihood ratio test for nested PR logistic regression models.
        Tests whether adding each additional set of predictors significantly improves
        model fit. Df: degrees of freedom added; Deviance: decrease in residual
        deviance; p-value: chi-squared test of improvement.") |>
  kable_styling(bootstrap_options = c("striped", "condensed"), full_width = FALSE)
tidy(glm_pr_main, conf.int = TRUE, exponentiate = TRUE) |>
  filter(term != "(Intercept)") |>
  mutate(
    term = case_when(
      term == "treatmentB" ~ "Treatment B vs A",
      term == "treatmentC" ~ "Treatment C vs A",
```

```r
      term == "sexMale"    ~ "Sex (Male vs Female)",
      .default = term
    ),
    across(c(estimate, conf.low, conf.high), ~ round(.x, 3)),
    p.value = formatC(p.value, format = "f", digits = 4)
  ) |>
  select(Term = term, OR = estimate,
         `95% CI low` = conf.low, `95% CI high` = conf.high,
         p = p.value) |>
  kable(caption = "Odds Ratios from Fitted PR Logistic Regression Model
        (Treatment + Age + Sex, n = 98). Treatment A is reference group. Bold row
        indicates dominant predictor. OR = odds ratio (exponentiated log-odds coefficient
        ); 95% CI = confidence interval; p = two-sided Wald test p-value.") |>
  kable_styling(bootstrap_options = c("striped", "condensed"), full_width = FALSE) |>
  row_spec(3, bold = TRUE)

par(mfrow = c(2, 2))
plot(lm_af_main, which = 1:4)
par(mfrow = c(1, 1))
vif_af <- vif(lm_af_main)

tibble(
  Predictor = names(vif_af[, 1]),
  `GVIF^(1/2Df)` = round(vif_af[, 3], 3)
) |>
  kable(booktabs = TRUE) |>
  kable_styling(latex_options = "hold_position", full_width = FALSE)
# Box-Tidwell
df_bt <- df_fitfr |>
  filter(!is.na(age), !is.na(sex)) |>
  mutate(age_log_age = age * log(age))
glm_bt  <- glm(PR ~ treatment + age + sex + age_log_age, data = df_bt, family = binomial)
bt_coef <- tidy(glm_bt) |> filter(term == "age_log_age")

# Hosmer-Lemeshow
hl_test <- hoslem.test(x = glm_pr_main$y, y = fitted(glm_pr_main), g = 10)

# VIF
vif_pr <- vif(glm_pr_main)

# Separation (min cell)
min_cell <- min(table(df_fitfr$treatment, df_fitfr$PR),
                table(df_fitfr$sex,       df_fitfr$PR))

tibble(
  Check = c("Box-Tidwell (age linearity)",
            "Hosmer-Lemeshow goodness-of-fit",
            "Complete separation",
            "VIF: treatment", "VIF: age", "VIF: sex"),
  Result = c(
    sprintf("beta = %.3f, p = %.3f", bt_coef$estimate, bt_coef$p.value),
    sprintf("chi-squared = %.2f, df = %d, p = %.3f",
            hl_test$statistic, hl_test$parameter, hl_test$p.value),
```

```r
    sprintf("No (min cell n = %d)", min_cell),
    sprintf("%.3f", vif_pr[1, 3]),
    sprintf("%.3f", vif_pr[2, 3]),
    sprintf("%.3f", vif_pr[3, 3])
  ),
  Conclusion = c(
    "Borderline; interpret with caution",
    "Adequate fit",
    "No separation",
    "No multicollinearity",
    "No multicollinearity",
    "No multicollinearity"
  )
) |>
  kable(booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position", "scale_down"), full_width = FALSE)
cooksd_pr    <- cooks.distance(glm_pr_main)
influential_pr <- which(cooksd_pr > 4 / nrow(df_fitfr))

tibble(obs = seq_along(cooksd_pr), cooksd = cooksd_pr,
       flagged = cooksd_pr > 4 / nrow(df_fitfr)) |>
  ggplot(aes(x = obs, y = cooksd, colour = flagged)) +
  geom_segment(aes(xend = obs, yend = 0), linewidth = 0.5) +
  geom_point(size = 2) +
  geom_hline(yintercept = 4 / nrow(df_fitfr), linetype = "dashed", colour = "grey40") +
  scale_colour_manual(values = c(`FALSE` = "#0D7680", `TRUE` = "#E8A838"),
                      guide = "none") +
  labs(x = "Observation index", y = "Cook's distance") +
  theme_bw(base_size = 11) +
  theme(panel.grid.minor = element_blank())
```

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